

External-FDT: Cloud-to-Building Control System Tunneling in BACnet Networks

Cameron Warren*
cwarre33@uncc.edu

Department of Computer Science
University of North Carolina at Charlotte

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Abstract

Building automation systems increasingly bridge private facility networks to public cloud infrastructure, creating undocumented attack surfaces. Through longitudinal analysis of internet-exposed BACnet Broadcast Management Devices (BBMDs), we discover a systemic pattern we term **External-FDT**: 71% of publicly visible BBMDs maintain Foreign Device Table entries pointing to public, routable IP addresses rather than internal network hosts. Of these, 42% tunnel to major cloud providers (AWS, Azure, DigitalOcean), while 25% reveal a novel **integrator concentration** phenomenon where a single external endpoint serves multiple geographically separated building networks. Our empirical analysis of 17 seeded BBMDs (longitudinally sampled over 47 days) characterizes this behavior and exposes potential supply-chain attack surfaces in building automation infrastructure. All findings derive from passive OSINT analysis of Shodan’s indexed banners—no active scanning of target systems.

1 Introduction

Building automation systems—responsible for HVAC, access control, fire safety, and critical facility operations—have historically relied on isolated broadcast networks. The BACnet protocol (ISO 16484-5)[?] enables these systems to span IP subnets through Broadcast Management Devices (BBMDs) that forward broadcast packets to registered Foreign Devices (FDs). These tunnels are typically confined to private networks[?].

We discover a systemic deviation from this model.

1.1 The External-FDT Phenomenon

Our analysis of internet-exposed BBMDs reveals that the majority maintain Foreign Device Table (FDT) entries pointing to **public, routable IP addresses**—creating undocumented bridges between building control networks and external infrastructure including major cloud providers (AWS, Azure, DigitalOcean).

This finding is distinct from previously characterized ICS exposure patterns. Prior work has focused on open ports[?], protocol vulnerabilities[?], and device enumeration[?], but no study has characterized *tunneling relationships* between building control systems and external infrastructure.

*ORCID: <https://orcid.org/0009-0007-4385-8778>

1.2 Key Contributions

This paper makes three empirical contributions:

1. **Characterization of External-FDT behavior:** We quantify the prevalence (71%) and patterns (cloud vs. integrator vs. unknown) of public-IP Foreign Device registrations in exposed BBMDs.
2. **Discovery of Integrator Concentration:** We identify a novel pattern where a single external endpoint serves multiple distinct building networks, suggesting supply-chain attack surfaces.
3. **Attribution Framework:** We validate passive OSINT methods for attributing external endpoints to cloud providers through IP-to-ASN mapping without active probing.

2 Background

2.1 BACnet and BBMDs

BACnet (Building Automation and Control Networks) is a standard protocol for intelligent building systems. BBMDs (Broadcast Management Devices) enable BACnet to span IP subnets by:[?]

1. Receiving broadcast messages on their local subnet
2. Forwarding encapsulated broadcasts to registered Foreign Devices
3. Receiving unicasts from Foreign Devices and re-broadcasting locally

A Foreign Device Table entry contains: IP address, port, TTL, and registration timestamp. Critically, the FD IP need not be on the same network as the BBMD— making cloud-to-building bridging architecturally possible.

2.2 Related Work

Prior ICS security research has focused on:

- Protocol vulnerabilities and specific CVE exploitation[?]
- Shodan-based exposure quantification[? ?]
- Modbus, DNP3, and BACnet object enumeration[?]

To our knowledge, no prior work has characterized *tunneling relationships* between internet-exposed building control systems and external infrastructure.

3 Methodology

3.1 Ethical Constraints

All analysis uses **passive OSINT only**: Shodan-indexed banners captured during normal internet scanning[?]. We performed:

- No active connection attempts to target IPs
- No BACnet Who-Is or Read-Property requests
- No authentication attempts
- No packet transmission to target networks

3.2 Data Source

Longitudinal Shodan dataset: 17 seeded BBMDs known to exhibit BACnet/WP fingerprinting, sampled March 4, 2026 through April 20, 2026 (47 days), enriched with Foreign Device Table extractions.

3.3 Classification Schema

We classify each Foreign Device Table entry:

Category & FD IP Range & Interpretation
Internal-FDT & RFC1918 (10/8, 172.16/12, 192.168/16) & Normal operation
External-FDT & Public routable & Requires investigation
Cloud & AWS, Azure, DO, GCP IP ranges & Cloud-hosted management
Integrator & Same FD serves multiple BBMDs & Third-party monitoring
Unknown & Uncategorized external & Further analysis needed

Table 1: FDT Entry Classification

Cloud attribution uses IP-to-ASN mapping against known cloud provider Autonomous System Numbers[?].

4 Results

4.1 Prevalence of External-FDT Behavior

Of 17 analyzed BBMDs:

- **12 (71%)** exhibited External-FDT behavior (public FD IPs)
- **5 (29%)** maintained only Internal-FDT (RFC1918) entries

This ratio suggests External-FDT is *systemic* rather than anomalous.

4.2 Cloud Provider Distribution

Among External-FDT cases ($N = 12$), destination infrastructure shows cloud concentration:

Provider & Count & Percentage
DigitalOcean & 4 & 33%
Amazon AWS & 2 & 17%
Microsoft Azure & 1 & 8%
Other / Unknown & 5 & 42%
Total Cloud & 5 & 42%

Table 2: Cloud Provider Attribution (External-FDT Cases)

DigitalOcean shows unexpected dominance—suggesting either preferred vendor relationships or lower barriers to entry for building automation integrators.

Attribute & Value
BBMD 1 IP & 66.58.248.125
BBMD 2 IP & 24.237.132.230
Shared FD IP & 216.67.73.166
Geographic separation & 4,000 km (different cities)
Observation window & 47 days
Port behavior & 13 rotating source ports each

Table 3: Integrator Concentration Pattern

4.3 Case Study: Integrator Concentration

The most significant finding: two distinct BBMDs share a common Foreign Device IP.

Implication: If the owner of 216.67.73.166 is a building automation integrator, this single endpoint bridges to at least two distinct client facility networks—suggesting a supply-chain attack surface where compromising one monitoring station yields access to multiple building control systems.

4.4 Persistence Analysis

Longitudinal data shows External-FDT relationships are stable:

- Average observation count: 47 scans per relationship
- Maximum: 930 observations over 47 days (single case)
- Port rotation patterns suggest NAT vs. direct connection

5 Discussion

5.1 Why This Matters

External-FDT creates three attack surfaces:

1. **Cloud compromise:** Breach of cloud-hosted management infrastructure yields direct access to building control networks
2. **Integrator pivot:** Single integrator compromise affects multiple facilities
3. **Exposure amplification:** Internet exposure of the FD endpoint compounds building network exposure

5.2 Limitations

Passive analysis cannot distinguish:

- Authorized SaaS integration vs. unauthorized bridging
- Active exploitation vs. dormant configuration
- Bidirectional tunnels vs. unidirectional monitoring

Attribution of cloud IPs to specific tenants or operators requires additional investigation.

6 Conclusion

We empirically demonstrate that External-FDT—cloud-to-building BACnet tunneling—is systemic in internet-exposed building automation infrastructure. The discovery of integrator concentration patterns suggests novel supply-chain attack surfaces requiring further research and coordinated disclosure.

Acknowledgments

This research was conducted using only publicly available Shodan index data. All facility identities and IP addresses have been anonymized except where already public in prior disclosures.